

BODY - SIDE VIEW

from the right · scale 0.94 : 1 · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
2 loops of #16 band
= 4 strands x 60 mm

BODY: black back
shell + white belly
cap (2 snap-fit parts)

2 M8 STEEL WASHERS
low at the back
→ CG low + behind

TAIL + Ø8 ROLLER
= 3rd wheel

HEEL WHEELS: Ø22, O-ring tyres,
0.5 mm eccentric hubs (C)

FEET hover 2 mm
(front stop)

PINION 10 T on the heel axle
(left side, under the crown)

TRIGGER PIVOT
+ crank (left
flipper, far side)

LOCK ROD in a
guide (see B)

CROWN 24 T hangs
on a pointed pin

BASE: oval tray,
wheel slots on top

FRONT VIEW (armed)

scale 0.64 : 1

rocks ±1.8° (C)

HEAD TURNS
on the neck
ring

TRIGGER = LEFT
flipper; raised
= ARMED (locked)

lock rod (B)

push it
down → GO

pinion 10 T
(toy's left)

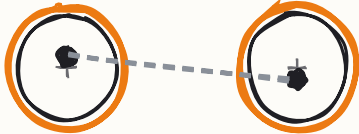
BASE: oval
tray, 42 wide

heel wheels
(track 32)

42 base

≈ 93 with the flipper raised

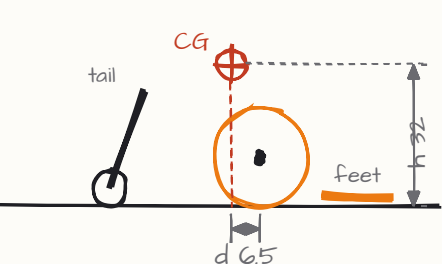
C WADDLE



LEFT wheel RIGHT wheel
axle hole 0.5 mm off-centre,
opposite sides (drawn x4)

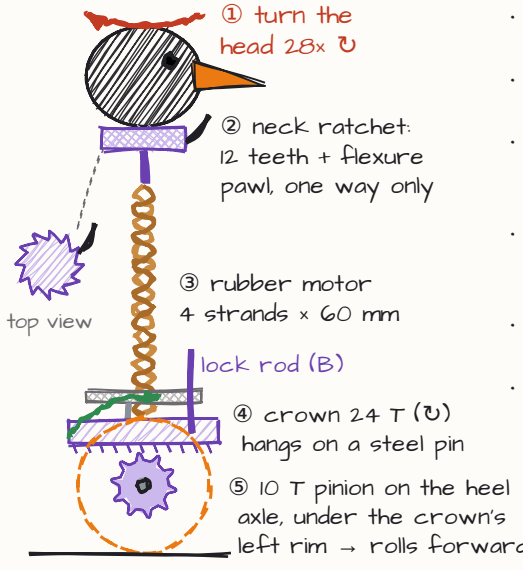
Roll = atan(1 + 32 mm track) ≈ ±1.8°
per wheel turn, the single tail roller
lets the body rock. It follows while
 $10\omega^2 < mg\text{track}/2$ ($1 \approx 8.5 \times 10^{-5}$
kgm²): below 0.55 m/s, i.e. at the
start and the last ≈ 1 m. Faster, the
heels hop 0.5 mm in turn (2.6 mJ/m
counted in the run model).

D HEELS + TAIL STANCE



Stands like an emperor: on its heels
(drive wheels) and stiff tail (roller);
the feet hover 2 mm as a front stop.
Launch $a = 25 \text{ mN} \div 44 \text{ g} \approx 0.6 \text{ m/s}^2$
just loads the tail - no wheelie.
Nose-dive needs $a = g/d \approx 2.0 \text{ m/s}^2$;
coasting slows at $< 0.5 \text{ m/s}^2$ ✓
Side tip needs 20°; waddle ±1.8° ✓

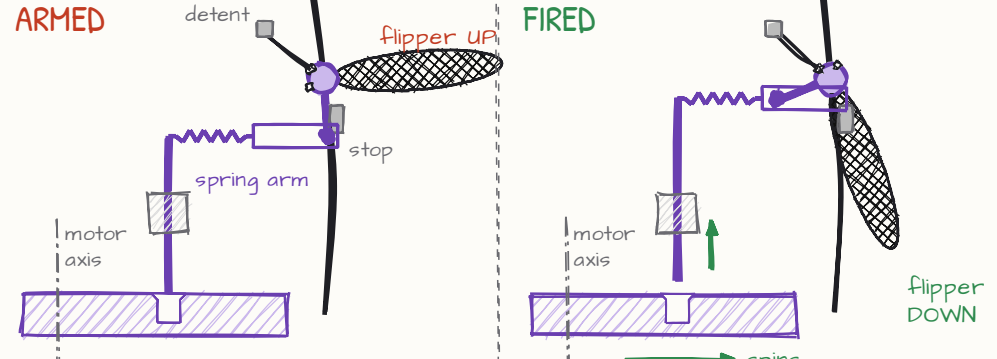
A ENERGY - TURN THE HEAD



head → ratchet → rubber → crown 24T → pinion 10T on the heel axle → wheels (x2.4)

- Rubber: 2 loops of #16 band = 4 strands x 60 mm (0.34 g), hook to hook
- 28 turns (≈ 1/3 of the ~90-turn break point) → 400 J/kg x 0.34 g ≈ 135 mJ
- 400 J/kg is our low assumption; we will measure torque vs turns (a 3rd loop adds 50 % if needed)
- Drag (Crr 0.015): rolling 6.5 + heel bores 6.2 + tail roller 4.5 = 17 mN; η 0.72 incl. the pin → 2.5 m needs ≈ 62 mJ (2.2x)
- Crown 24 : pinion 10 → 2.4 wheel turns per motor turn, 28 turns = 4.6 m
- Time-step model (torque ∝ turns left, waddle hops): run ≈ 4.6 m, top 1.0 m/s; worst case (250 J/kg) 3.1 m. Unwound, the head free-spins on its ratchet as it coasts.

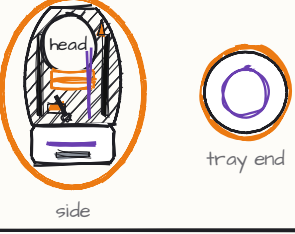
B TRIGGER - FLIPPER CRANK LOCK



UP: detent 3 N·mm (15x flipper weight), rod 2.5 mm into a 3 mm straight bore

light press 0.07 N (tipping needs 5 N·mm); 2nd notch holds it, rod 4 mm up → GO

E PACKS INTO THE EGG



Scale 1 : 4. The 2 body parts close into a hull (61 tall with the open neck, 46 x 50) holding the head, beak, wheels, flippers, tail + roller, feet, rod, pinion, axles and bands; the oval tray (crown, washers, pin inside) sits in the bottom of the egg. Tightest: tray 21.4 of 23.3 mm; body top 4.5 mm clear. Assembled 109 mm tall → bigger than the egg ✓

BIOMIMICRY



Emperor penguins rest on their heels and stiff tail with toes lifted, and waddle side to side. Copied: heel drive wheels + tail roller (3-point stance), low CG, rocking wheels. Bonus: the head swivels (it is the winding key).

HOW IT PLAYS

- 1 Raise the LEFT flipper - click: the lock rod springs onto the crown.
- 2 Turn the head 28x clockwise (click-click...), stop beak-forward.
- 3 Set it down - it stands on its heels and tail, fully wound.
- 4 Press the flipper down lightly → it waddles off, ≈ 4.6 m.

vs Concept 1: upright body, twisted rubber motor (turn key), flipper trigger.

NUMBERS

- Mass ≈ 44 g, ≈ 86 % printed PLA
- CG 32 mm high, 6.5 mm behind the heel axle (81 % on the heels)
- Motor ≈ 135 mJ; 2.5 m needs ≈ 62 mJ
- Gears ml, printed: crown 24 T → pinion 10 T (x2.4)
- Run ≈ 4.6 m (worst case 3.1 m), top speed ≈ 1.0 m/s
- Lock-rod side load ≤ 0.17 N (1.54 N·mm ÷ 9.2 mm)
- Bought (9): 2 axles + 1 pivot pin (steel), 2 O-rings, 2 bands, 2 washers

RULES CHECK

- ✓ ≥ 75 % printed PLA (≈ 86 %)
- ✓ All parts fit the egg (box E); the built toy is bigger
- ✓ Holds its energy until the flipper is pushed down
- ✓ Trigger is a flipper crank - no removable pin, stays together
- ✓ No energy stored when apart (rubber untwisted)
- ✓ Snap / press fits: no glue, no tools
- ✓ Gears are our printed parts, not commercial toy parts

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 2 · sheet 1 of 1

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - emperor stance + waddle
Toy type: Running toy (≥ 2.5 m)
Scale / units: mm · side view 0.94 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____